

Problem

Evaluate the following complex numbers:

a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]^*$

b) $\frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$

Step-by-step solution

Step 1 of 6

(a)

Consider the following complex equation.

$$\left([(5 + j2)(-1 + j4)] - [5\angle 60^\circ] \right)^*$$

Determine the rectangular form.

$$\left([(5 + j2)(-1 + j4)] - [5\angle 60^\circ] \right)^*$$

$$[-5 + j20 - j2 + j^2 8]^* - [5\angle 60^\circ]^*$$

Consider $j^2 = -1$.

$$[-5 + j18 - 8]^* - [5\angle 60^\circ]^*$$

$$[-13 + j18]^* - [5\angle 60^\circ]^* \dots\dots (1)$$

Step 2 of 6

Convert the polar form into rectangular form.

$$5\angle 60^\circ = 5(\cos 60^\circ + j \sin 60^\circ)$$

$$5\angle 60^\circ = 5\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$$

$$= 2.5(1 + j\sqrt{3})$$

$$= 2.5 + j2.5(\sqrt{3})$$

$$= 2.5 + j2.5(1.7321)$$

$$5\angle 60^\circ = 2.5 + j4.3303 \dots\dots (2)$$

Step 3 of 6

Substitute equation 2 in equation 1.

Consider $[-13 + j18]^* = [-13 - j18]$ and $[2.5 + j4.3303]^* = [2.5 - j4.3303]$.

$$\begin{aligned} [-13 + j18]^* - ([2.5 + j4.3303]^*) &= -13 - j18 - (2.5 - j4.3303) \\ &= -13 - j18 - 2.5 + j4.3303 \\ &= -15.5 - j13.67 \end{aligned}$$

Therefore, the rectangular form is $\boxed{-15.5 - j13.67}$.

Step 4 of 6

(b)

Determine the rectangular form.

Consider the following complex equation.

$$\frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 \dots\dots (3)$$

Convert the polar form into rectangular form.

$$\begin{aligned} 3\angle 40^\circ &= 3(\cos 40^\circ + j \sin 40^\circ) \\ &= 3(0.766 + j0.643) \end{aligned}$$

$$3\angle 40^\circ = 2.298 + j1.93 \dots\dots (4)$$

Consider $10\angle 30^\circ$.

$$\begin{aligned} 10\angle 30^\circ &= 10(\cos 30^\circ + j \sin 30^\circ) \\ &= 10(0.866 + j0.5) \end{aligned}$$

$$10\angle 30^\circ = 8.66 + j5 \dots\dots (5)$$

Step 5 of 6

Substitute equation 4 and 5 in equation 3.

$$\begin{aligned} \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 &= \frac{10 + j5 + (2.298 + j1.93)}{-3 + j4} + (8.66 + j5) + j5 \\ &= \frac{12.298 + j6.93}{-3 + j4} + 8.66 + j10 \\ &= \frac{12.298 + j6.93 + (8.66 + j10)(-3 + j4)}{(-3 + j4)} \\ &= \frac{12.298 + j6.93 + (-25.98 + j34.64 - j30 + j^2 40)}{-3 + j4} \\ \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5 &= \frac{12.298 + j6.93 + (-25.98 + j4.64 - 40)}{-3 + j4} \\ &= \frac{12.298 + j6.93 - 65.98 + j4.64}{-3 + j4} \\ &= \frac{-53.682 + j11.57}{-3 + j4} \\ &= 10.983\angle 40.97 \end{aligned}$$

Step 6 of 6

Convert $10.983\angle 40.97$ into rectangular form.

$$\begin{aligned} 10.943\angle 40.97 &= 10.983(\cos 40.97 + j \sin 40.97) \\ &= 10.983(0.755053 + j0.6557) \\ &= 8.293 + j7.2 \end{aligned}$$

Therefore, the rectangular form is $8.293 + j7.2$.